

Apex vs Pinnacle — Four-Season Evidence

Out-of-sample goal-forecast accuracy vs the closing market, 2022/23 – 2025/26

Across 1,283 Premier League matches, Apex reduced goal-prediction error by 8.4% versus the goal expectations implied by Pinnacle's closing markets — outperforming the market in every one of the four seasons tested.

A repeatable forecasting edge over one of the sharpest books in the world, measured out-of-sample, match by match.

Headline

- **8.4% lower** mean absolute error across 1,283 matches (Apex 0.937 vs 1.023).
- **0.086 fewer goals** of forecast error per team, per match, on average.
- **Lower error in all four seasons** tested.
- Apex was the closer forecast in **55.8%** of individual matches.

Season by season

Season	Matches	Apex MAE	Pinnacle MAE	Error reduction	Apex closer	z
2022/23	340	0.957	0.999	4.2%	51.5%	0.54
2023/24	336	0.976	1.117	12.6%	55.7%	2.07
2024/25	305	0.918	1.010	9.1%	59.7%	3.38
2025/26	302	0.890	0.958	7.1%	57.0%	2.42
All four	1,283	0.937	1.023	8.4%	55.8%	4.16

MAE = mean absolute error in goals per team per match (lower is better). Error reduction = (Pinnacle MAE – Apex MAE) / Pinnacle MAE. Apex closer = share of matches where Apex's total-goal error was smaller.

Honest read: Apex's average error is lower in all four seasons (error reduction 4.2–12.6%), by an average of 0.086 goals per team per match. 2022/23 was a near-tie match-by-match (51.5%) but still produced the lower average error. The edge is positive in every season; it does not strengthen monotonically — the strongest year was 2023/24, the weakest the first (2022/23).

Why this is credible

1. Out-of-sample (walk-forward). For every matchday the model is fit only on matches that finished *before* that day, then predicts forward — it never sees a result before predicting it (min. 150 prior matches). Source: `apex_v1_backtest_bic.py`.

2. Benchmark. Pinnacle publishes no expected goals, so they are extracted from the *closing* lines: the Over/Under line as implied expected total, the Asian Handicap line as implied expected supremacy, decomposed per team:

```
pinnacle_home = (close_OU + (-AH_line)) / 2
pinnacle_away = (close_OU - (-AH_line)) / 2
```

The closing line is the market's balance point — a standard, close proxy for its mean expectation.

3. Significance — two tests. *Win-rate:* 55.8% of matches closer vs a 50/50 null gives $z = 4.16$, two-tailed $p = 3.2 \times 10^{-5}$. *Paired error test (stronger):* the per-match error difference (Apex – Pinnacle) vs zero gives mean -0.086 goals, $t = -7.1$, $p \approx 1 \times 10^{-12}$ — the error reduction itself, not just the win rate, is highly significant.

4. Full sample. Every match with a Pinnacle closing line is included — no cherry-picking.

The claim

The evidence shows Apex predicted match goals more accurately than Pinnacle's closing-market expectations across four consecutive Premier League seasons. This measures **forecasting accuracy** — the right claim for FPL projections and goal-line analysis. It does **not** claim a betting-profit edge (a separate, unproven question).

“Across 1,283 Premier League matches, Apex reduced goal-prediction error by 8.4% versus Pinnacle's closing-market expectations, outperforming the market in every season tested.”

Audit trail: per-match data for all 1,283 fixtures (actual score, Apex goals, Pinnacle goals, both errors, who was closer) in `apex_evidence_all_seasons.csv` — downloadable so any third party can reproduce every figure independently. Source: `apex_v1_backtest_bic.py` (walk-forward).